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AI-driven virtual cell models in preclinical research: technical pathways, validation mechanisms, and clinical translation potential

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AI-driven virtual cell models show the potential to transform the paradigm of life sciences research by integrating multimodal omics data (e.g., single-cell transcriptomics and proteomics) with advanced algorithms such as deep generative models and graph neural networks to enable high-precision predictions of drug responses, gene perturbations, and disease progression. These models enable high-precision predictions of drug responses, gene perturbations, and disease progression. This review outlines the technical pathways and validation mechanisms of virtual cells, emphasizing a closed-loop workflow from computational evaluation to experimental verification using CRISPR assays and organoid platforms. The applications of virtual cells in personalized drug screening and disease modeling are highlighted, showcasing their potential to reduce animal testing and optimize therapy. However, challenges in regulatory acceptance, data privacy, and model interpretability remain. Global policy and standardization trends are driving clinical translation, and future advancements will involve cross-disciplinary integration and greater standardization to enhance the impact of virtual cells in precision medicine and drug discovery.

AI-driven virtual cell models present a paradigm-shifting prospect for life-science research by supporting mechanistic, predictive simulation of functional responses^{1–3}. A “Virtual Cell” is a computational model that simulates cellular functional states, signaling networks, and their dynamics under diverse perturbations^{1,4}. Compared with conventional computational cell models, AI-driven virtual cells learn latent patterns from large-scale, multimodal biological data to construct predictive cellular state spaces^{5,6}. Terminological fragmentation across the literature—spanning “virtual cell,” “digital cell,” and “digital twin”—drives our decision to foreground “virtual cell” as the field’s integrative label.

The emergence of this concept has been enabled by the rapid advancement of single-cell RNA sequencing (scRNA-seq), spatial transcriptomics (ST), proteomics, and other high-throughput omics, together with the introduction of novel Artificial Intelligence (AI) architectures such as deep generative models, graph neural networks, and physics-constrained neural networks^{7,8}. By integrating multi-omics data from diverse sources, virtual cell models can reconstruct functional networks and signaling

pathways across multiple scales, including the subcellular, single-cell, and cell-population levels¹.

In preclinical drug research before Investigational New Drug (IND) submission, virtual cell models already demonstrate diverse applications: by simulating gene knockout, overexpression, or mutation, they predict functional roles in disease progression and, in combination with CRISPR validation, help screen potential drug targets⁹. In addition, these models can perform high-throughput *in silico* efficacy predictions across large candidate libraries to assess selectivity and potential adverse effects^{6,10}. During lead optimization, simulations of absorption, distribution, and metabolism provide predictions that more closely approximate human physiological contexts^{11,12}. Proof-of-concept, a multimodal single-cell virtual-cell model reconstructs mouse pancreatic ontogeny and validates NEUROD2-dependent ϵ -cell differentiation¹³; at scale, State—trained on $>10^8$ cells—yields superior predictions of drug-response perturbations across cell types¹⁴. Emerging practical utility of virtual cell models for mechanistic discovery and drug screening.

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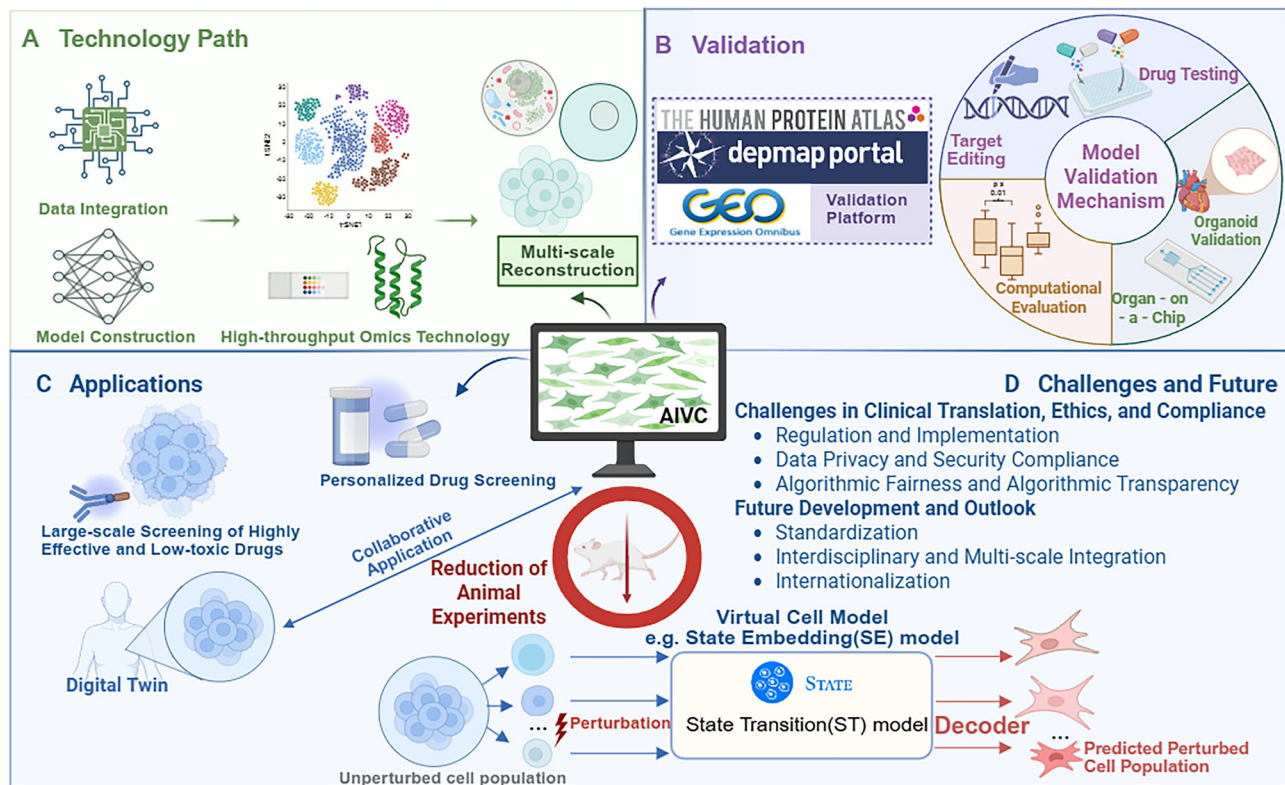


Fig. 1 | Comprehensive Overview of AI-Driven Virtual Cell Models in Preclinical Research. This figure systematically illustrates the entire lifecycle of AI-driven virtual cell models in preclinical research, encompassing technical pathways, validation mechanisms, application scenarios, as well as challenges and future directions in clinical translation. **A Technical Pathways** Multimodal omics data (e.g., single-cell transcriptomics, spatial transcriptomics, proteomics) are integrated via advanced AI techniques—including deep generative models (e.g., UNAGI, scDiffusion), graph neural networks (e.g., DrugCell-GNN, GraphST), and physics-informed neural networks (e.g., BioPINN, VCBA)—to reconstruct cellular networks at subcellular, single-cell, and cell-population scales. Specialized platforms such as VCell (molecular/cellular level), PhysiCell (tissue level), and Chaste (organ level) enable multi-scale modeling, while foundation models like GeneFormer and State facilitate cross-level knowledge transfer from the gene/cell level to the organ and patient levels. **B Validation Mechanisms** A two-tier validation framework is adopted: Computational Evaluation: Metrics such as distributional concordance, prediction accuracy, and uncertainty quantification are assessed using platforms like DepMap Portal and GEO. Experimental Validation: Predictions are verified via CRISPR assays, organoids (e.g., patient-derived intestinal organoids for toxicity testing), and organ-on-chip systems (e.g., hiPSC-CMs for cardiotoxicity prediction). For instance, a virtual cell model based on hiPSC-CMs accurately predicted doxorubicin-induced

cardiotoxicity, consistent with clinical observations; VCBA-simulated hepatocyte mitochondrial toxicity matched in vitro experimental results. A closed-loop iteration of “computation→experiment→translation” ensures model robustness. **C Application Scenarios** Virtual cell models have diverse applications in preclinical research: Precision Drug Screening: Large-scale in silico screening of low-toxicity and high-efficacy compounds (e.g., million-compound virtual screening by the State model) and personalized drug response prediction (e.g., rare disease modeling integrated with digital twins). Mechanistic Inference: Elucidation of regulatory pathways (e.g., the role of NEUROD2 in pancreatic cell differentiation) and dynamic processes of drug-cell interactions. Reduction of Animal Experiments: Replacement of traditional animal models in toxicity and efficacy assessments (e.g., organoid-based antibody toxicity testing consistent with clinical reports). **D Challenges and Future Directions in Clinical Translation** Key hurdles include regulatory ambiguity (e.g., FDA’s case-by-case IND review), data privacy (addressed by differential privacy and federated learning), intellectual property disputes, and algorithmic bias (mitigated by explainable AI tools like SHAP). Future developments will focus on standardization (e.g., SBML/CellML for model interoperability), interdisciplinary multi-scale integration (e.g., fusion of organoids, virtual cells, and digital twins), and international collaboration to advance clinical application. (Created in <https://BioRender.com>).

Virtual cell models also help elucidate tissue-specific differences in drug responses¹⁵. For example, integrating multi-omics data from hepatocytes and renal tubular epithelial cells enables prediction of tissue-specific toxicological responses under the same drug exposure⁶. This not only helps explain interindividual variability in adverse drug reactions but also offers a technical pathway to reduce animal experimentation and advance New Approach Methodologies (NAMs)^{6,16}.

The advent of virtual cell models is shifting biomedicine from a wet-lab-dependent, “validation-driven” paradigm toward a data- and prediction-driven simulation-validation closed loop¹⁶. In sum, AI-driven virtual cell models are becoming a key bridge between molecular-mechanism research and preclinical drug evaluation, providing data-driven support throughout the pipeline from hypothesis generation to mechanistic verification¹⁵. Driven by an evidence-progression framework, the narrative advances from model construction to evaluation and validation, to virtual-cell use cases, translational barriers, and forward outlook.

This translational driver integrates technical capability, testability, utility, and compliance pathways into a preclinical evidence chain (Fig. 1).

Technical Pathways

Precision virtual-cell construction sits at the forefront of the biology–AI nexus. Its core task is to integrate multimodal data, especially single-cell and spatial omics, thereby underpinning model development. Driven by complementary learners, deep generative models sculpt the cellular state space, graph neural networks capture intercellular interactions, and physics-informed neural networks embed biological laws, collectively shaping a virtual-cell model that delivers accurate predictions with defensible mechanistic interpretability.

Multimodal data integration

A primary challenge in building high-precision virtual cell models is integrating heterogeneous biological modalities, including scRNA-seq, ST,

Table 1 | Key data sources for virtual-cell modeling and typical uses

Data modality	Representative platforms / repositories	Resolution / coverage	Typical modeling uses	Notes
Single-cell transcriptomics (scRNA-seq)	NCBI GEO; EBI ArrayExpress; Human Cell Atlas (HCA); HuBMAP; CZ CELLxGENE	Cell level; transcript abundance	Cell-type identification; developmental/disease trajectory inference; perturbation-response modeling	Primary archives provide the most comprehensive raw datasets. Curated atlases supply uniformly processed, analysis-ready data that help mitigate batch effects.
Spatial transcriptomics (ST)	NCBI GEO; STOmicsDB; SpatialDB; HCA; HuBMAP	Subcellular to multi-cellular spots; spatial gene expression	Spatial deconvolution; tissue microenvironment inference; cell–cell interaction modeling	The data ecosystem is maturing rapidly. Specialized databases offer visualization tools, but update cadence requires attention. Primary archives are common deposition sites, though standardized spatial metadata may be incomplete.
Multi-omics (e.g., scATAC-seq)	NCBI GEO; EBI ArrayExpress; HCA; HuBMAP; CZ CELLxGENE	Cell level; joint RNA and chromatin accessibility	Cross-modal alignment; gene regulatory network inference; lineage tracing; fine-grained state delineation	Integrated atlases are critical for discovering joint distributions and building multi-layered cellular models.
Proteomics (mass spectrometry)	ProteomeXchange (PX) Consortium (e.g., PRIDE, MassIVE)	Protein level; abundance and post-translational modifications (PTMs)	Fusion with transcriptomics to analyze functional states; modeling pathway activity	PX supplies unified global standards for submission and retrieval of raw and processed MS data, ensuring integrity and reusability.
Proteomics (immunolocalization)	Human Protein Atlas	Subcellular to tissue level; protein spatial localization	Constraining protein placement in spatial cell models; validating gene-expression patterns	Microscopy-based, spatially resolved protein evidence from HPA is a key complement to quantitative MS proteomics.

epigenomics, proteomics, and metabolomics^{1,2}. Deep generative models and graph neural networks show strong potential for simulating single-cell states and predicting drug responses¹⁷, while physics-informed neural networks (PINNs) can enhance interpretability and biological plausibility¹⁶.

scRNA-seq resolves gene expression at single-cell resolution but loses spatial context^{18,19}; ST preserves spatial structure yet faces limits in resolution, throughput, and gene coverage^{20,21}. High-throughput imaging-driven advances in spatial transcriptomics now enable the simultaneous detection of ~20,000 genes, producing near-whole-transcriptome spatial maps and progressively mitigating the historical gene-coverage bottleneck of imaging-based ST^{22,23}.

To bridge these gaps, AI-based cross-modal fusion methods have been proposed^{17,8,24–27}. For instance, SpatialScope uses deep generative modeling to project high-dimensional scRNA-seq data into ST coordinates, enhancing resolution and imputing missing genes⁷. VAE (variational autoencoder)-based frameworks such as scVI and totalVI integrate multi-batch single-cell data and fuse transcriptome with proteome for joint denoising and analysis^{8,28}. Domain adaptation reduces cross-platform bias: scAdapt employs adversarial training to align distributions between ST and scRNA-seq, improving spatial deconvolution accuracy^{7,24}; conditional normalization and attention help detect rare cell types and low-abundance genes^{25,29}. On high-dimensional integration, MultiVI maps transcriptomic and epigenomic modalities into a shared latent space to improve cell-type classification, trajectory inference, and lineage analysis²⁶; GraphST couples graph structures with expression and spatial proximity to drive cell-type identification and spatial relationship reconstruction²⁷.

Recent advances have led to a new wave of methods for integrating multi-omics data. For example, the GLUE (graph-linked unified embedding) model incorporates prior regulatory networks to project heterogeneous single-cell omics into a shared embedding space, achieving strong cross-species integration across millions of cells³⁰. Driven by optimal-transport-based coupling across modalities, the moscot framework collectively shapes single-cell multi-omics into coherent spatiotemporal developmental trajectories in the mouse, with its model-predicted key regulators validated experimentally¹³. Moreover, a vision has been articulated for multimodal foundation models: that is, pretrained unified models that integrate genomics, transcriptomics, epigenomics, proteomics, metabolomics, and spatial omics to comprehensively characterize cellular states³¹. Such models would enable context-specific transfer learning for panoramic single-cell atlasing.

These strategies enable virtual cells to integrate multilayer omics more comprehensively, forming richer, more accurate inputs for downstream drug response prediction, perturbation simulation, and mechanism inference^{1,2}.

Large public databases provide critical training data^{32–34}: TCGA (The Cancer Genome Atlas) for tumor genomics/transcriptomics^{35,36}; the Human Protein Atlas (HPA) for tissue/cell-level protein localization^{37–39}; and GEO (Gene Expression Omnibus) for diverse omics datasets^{40–42}. All harbor inherent biases^{43–46}. Corrective techniques help: ComBat, Harmony, and related methods attenuate batch effects^{47–49}; domain adaptation (e.g., scAdapt) aligns distributions within models⁵⁰. Rigorous normalization and correction bolster generalization and reduce misleading predictions driven by data bias^{46,51–54} (see Table 1).

Omics foundation models and cross-hierarchical transfer

Foundation models for virtual cells trained on tens of millions of single cells deliver strong performance in cell-type annotation and gene-function inference (GeneFormer⁵⁵; scFoundation⁵⁶). Meanwhile, models that target complex genetic perturbations: GEARS⁵⁷ and the Arc Institute's State model⁵⁸, exhibit measurable generalization across cell types. Driven by cross-scale knowledge transfer, the next critical task is to migrate capabilities learned at the gene/cell level to the patient level, where explicit modeling of tissue microenvironments, pharmacokinetics, and other systems constraints becomes indispensable. To this end, several studies use transfer learning to achieve cross-scale prediction: CODE-AE extends drug-response prediction from cell lines to patient tumor transcriptomes⁵⁹, and scDEAL integrates population-level and single-cell data to improve individualized drug-sensitivity prediction⁶⁰. CSG2A, by leveraging condition-specific gene and gene-to-gene attention, pretrains on LINCS L1000 and fine-tunes on GDSC to transfer perturbation knowledge from the gene level to drug-response prediction in cells and patients, enhancing the interpretability of virtual-cell models⁶¹. Together, these advances suggest that elevating gene-level pretraining to organ- and patient-level virtual models is feasible, provided that additional determinants—tissue context, ADME (absorption, distribution, metabolism, and excretion) processes, and related factors—are rigorously incorporated.

Deep generative models

Deep generative models are central to constructing predictive and generative cellular state spaces^{1,17}. Beyond fitting high-dimensional data

Table 2 | Common AI methods for virtual cell: tasks, I/O, and availability

Method	Task	Input	Output	Validation strength	Open-source / code / web
scVI / totalVI	Batch integration, denoising, joint RNA–protein analysis	scRNA-seq ($\pm$ protein)	Latent space	Multi-dataset benchmarks; cross-laboratory validation	https://github.com/scverse/scvi-tools (scVI/totalVI)
GLUE / MultiVI	Cross-modal fusion and alignment	RNA + ATAC ($\pm$ spatial transcriptomics)	Shared representation space	Systematic evaluation on public datasets	https://github.com/gao-lab/GLUE (GLUE); https://github.com/scverse/scvi-tools (MultiVI)
GraphST / SpaGCN	Driven by spatial adjacency graphs, model tissue architecture and microenvironment niches	Expression matrix + spatial adjacency	Cell type / niche	Validation on tissue benchmark datasets	https://github.com/JinmiaoChenLab/GraphST (GraphST); https://github.com/jianhuupenn/SpaGCN (SpaGCN)
CPA / GEARS	Perturbation-effect prediction	Baseline RNA + perturbation condition	Δ state	Leave-one-drug/gene generalization tests	https://github.com/theislab/cpa (CPA); https://github.com/snap-stanford/GEARS (GEARS)
DrugCell-GNN	Predict drug efficacy and synergy	Cell transcriptome + drug molecular graph	Drug response	External drug screens and independent dataset validation	https://github.com/idekerlab/DrugCell (DrugCell)
GeneFormer / scFoundation / STATE	Driven by large-scale pretraining, learn transferable cell representations	Large-scale RNA corpora	Transferable cell representations	Transfer validation across diverse downstream tasks	https://github.com/kjobject/geneformer (GeneFormer); https://github.com/biomap-research/scFoundation (scFoundation); https://github.com/Arclnstitute/state (STATE)

distributions, they can generate synthetic expression profiles in latent space for virtual experiments under drug perturbations or genetic edits^{5,17}. Recent work has simulated disease trajectories and drug responses; for example, the graph-structured VAE-GAN model UNAGI captured single-cell dynamics in idiopathic pulmonary fibrosis and predicted potential anti-fibrotic activity of nifedipine, validated in patient lung tissues and proteomics⁶². Beyond classic frameworks such as VAEs, flow matching and diffusion models now define a new paradigm in generative modeling. Flow matching directly learns continuous-time transformations between data distributions to enable efficient generation of high-dimensional, structured data⁶³; diffusion models formalize noise injection and denoising inversion with stochastic differential equations, progressively approaching the target distribution, and have achieved notable advances in image generation and single-cell data simulation⁶⁴.)Driven by diffusion dynamics coupled with a pretrained foundation model, scDiffusion generates single-cell transcriptomes with high fidelity and strong diversity^{65,66}. Pressure for cross-cell-type generalization drives model design: integrating perturbation data across lineages and enforcing priors from biological networks, mechanistically improves transfer to unseen perturbations. Against this backdrop, current deep learning approaches, in aggregate, do not surpass simple linear baselines by a large margin⁶⁷, whereas a knowledge-graph-guided framework (GEARS) reports ~40% accuracy improvement for predicting transcriptional responses to multi-gene perturbations⁵⁷. Mechanism-anchored validation remains sparse; only a few virtual cell predictions have been confirmed via targeted interventions that verify model-nominated key regulators¹³. Establishing standardized experimental validation pipelines should therefore be elevated as a field-level priority. Thus, generative models provide a powerful *in silico* experimental platform for hypothesis testing and therapeutic exploration¹ (see Table 2).

Graph neural networks

Cells influence each other through signaling pathways, cell–cell communication networks, and spatial proximity^{2,29}. Biologically faithful virtual cells must model such topology and information flow¹. Graph neural networks (GNNs) naturally suit cell-relationship modeling in single-cell omics²⁹. Treating each cell as a node, GNNs iteratively aggregate neighborhood information to capture contextual dependencies^{68–70}. scGNN embeds scRNA-seq into a graph and applies multi-layer GCNs for feature aggregation, outperforming t-SNE (t-distributed Stochastic Neighbor

Embedding)/UMAP (Uniform Manifold Approximation and Projection) for cell-type identification, imputation, and trajectory inference²⁹.

For multimodal fusion, PINNACLE integrates scRNA-seq with protein–protein interaction networks via graph attention to better detect rare subpopulations and regulatory patterns^{71–73}, aiding immune-cell subtyping and resistance mechanisms⁶. GNNs also extend to drug response prediction^{74–76}. DrugCell-GNN unifies cellular transcriptomes, drug molecular graphs, and known target networks to predict sensitivity and synergy, improving accuracy in anticancer screening^{1,10}. With the rise of ST, SpaGCN couples spatial adjacency with expression matrices to resolve cell types and tissue organization, excelling in tumor microenvironment heterogeneity analyses⁷.

Overall, GNNs endow virtual cells with multiscale modeling, from single-cell regulatory patterns to population-level coordination^{1,2,77}. Integrating ST and live-cell imaging will further sharpen tissue-microenvironment realism and empower disease-mechanism and drug-development studies^{6,7}.

Physics-informed neural networks

Although deep learning models demonstrate powerful data fitting and generation capabilities in virtual cell simulations, most are “black-box” models that lack explicit expression and constraints of known biological laws. The core idea of Physics-Informed Neural Networks is to incorporate known biophysical laws, dynamic equations, or constraints during the model training process to ensure that the predicted results are biologically plausible^{16,78–82}. This approach combines theoretical models with data-driven deep networks, balancing flexibility and interpretability¹⁶.

In drug metabolism and toxicology simulations, the Virtual Cell-Based Assay (VCBA) platform combines organelle-level dynamic models (e.g., mitochondrial membrane potential changes, reactive oxygen species production) with dose-time integral models to simulate dynamic responses of hepatocytes and cardiomyocytes under drug exposure^{9,11}. These models have demonstrated high consistency with some *in vitro* experimental data and clinical adverse event reports in predicting drug-induced liver injury (DILI) and cardiotoxicity^{9,12}. Hybrid constraint methods are also a current trend^{83–85}. For example, integrating the Flux Balance Analysis (FBA) model of metabolic pathways with neural networks can improve prediction accuracy for environmental disturbance responses while maintaining metabolic flux conservation⁸⁴. Similarly, embedding the classic Hodgkin–Huxley equation of ion channel electrophysiology as a prior in

cardiomyocyte models can more accurately predict drug-induced action potential changes^{85,86}.

However, constructing PINNs still presents challenges^{1,6,16,87,88}. Predictive improvement in PINNs hinges on the completion of biophysical knowledge and the identification of kinetic parameters, rather than hardware scaling; incomplete laws and parameter gaps impose a hard ceiling. Mathematical descriptions of biological systems are often incomplete, and some dynamic parameters are difficult to obtain¹⁶. The introduction of physical constraints increases the complexity of model training and sometimes conflicts with data fitting objectives, resulting in convergence difficulties¹. Furthermore, in a multi-scale, multi-modal data environment, selecting and weighting constraints appropriately is a key challenge in model design^{6,16}. With the improvement of computational hardware, PINNs are expected to become an important component of virtual cell models in the future^{1,6,16}.

The various technical approaches mentioned above each have their own advantages, and in virtual cell modeling practices, they are not mutually exclusive but can work synergistically¹. Currently, the integration of multimodal data ensures comprehensive and high-quality input for the models^{89,90}. Next, deep generative models are used to establish baseline predictive capabilities for cell state spaces, such as generating cell transcriptomes under unseen perturbation conditions^{17,91}. Furthermore, a graph neural network module is introduced, and previous research has explored combining generative models, graph networks, and physical constraints to build more powerful integrated models^{62,92,93}. Despite the limitations of different methods, the comprehensive application of multiple techniques is expected to construct a more comprehensive and accurate virtual cell model¹.

Platforms and toolboxes at a glance

A compact landscape of virtual-cell and cell-scale modeling platforms now anchors the fourth technical route, spanning molecular networks, multicellular systems, and organ-level physiology (see Table 1). These platforms can be broadly grouped by their core modeling targets and methodological foundations, each serving distinct research questions.

Driven by mechanistic representations of intracellular processes, several toolchains target gene regulation, signal transduction, and related pathways. VCell (Virtual Cell) provides a unified cell-biology environment that supports spatially resolved and well-mixed (nonspatial), deterministic and stochastic modeling, and it natively integrates image-analysis workflows^{94,95}. For quantitative kinetics, COPASI (Complex Pathway Simulator) specializes in ODE/SDE (stochastic differential equation)-based biochemical network modeling and parameter estimation^{96–99}. To address combinatorial explosion in rule-based signaling models, BioNetGen (BNG)^{100–102} and PySB¹⁰³ enable compact specification of reaction rules and efficient state-space handling. Logic-network frameworks such as CellNOpt^{104–106} and the Cell Collective¹⁰⁷ infer apoptosis signaling circuitry from perturbation datasets. Application-oriented platforms like VCBA focus on environmental toxicology and biokinetic simulation^{9,108,109}.

At the multicellular and tissue scale, agent-based modeling (ABM) is widely used to capture population behaviors and tissue repair dynamics. PhysiCell emphasizes 3D, tissue-scale ABM and is particularly suited for simulating the tumor microenvironment^{110–113}. Guided by cell–cell mechanics and morphodynamics, CompuCell3D^{114–117} and Morpheus^{118,119} implement Cellular Potts/GGH frameworks that couple cell mechanics with reaction–diffusion processes.

Specialized engines and emerging AI tools further extend virtual-cell modeling. At the subcellular scale, Smoldyn^{120,121}, MCell¹⁰², and ReaDDy^{122,123} perform particle-based stochastic reaction–diffusion, enabling molecular-level transport and interaction studies. For organ-system modeling, Chaste¹²⁴, OpenCOR/CellML (Cell Markup Language)^{125,126}, and OpenSim/Physiome^{125,127} support multiscale cardiac electrophysiology and biomechanics. Recently, AI-driven platforms have gained traction: DeepCell¹²⁸ and CellPose^{129–131} deliver high-accuracy cell image segmentation and analysis, while NVIDIA Modulus¹³² provides a

framework for physics-constrained neural networks—directly aligning with the three advanced AI routes discussed above (see Table 3) (Fig. 2).

Validation Mechanisms

Collective shaping by a closed-loop validation architecture drives virtual cell models from methodological novelty to actionable biology. At its core is a systematized pipeline coupling computational assessment (fit, stability, uncertainty) with experimental corroboration, iterated in cycles. This mechanism foregrounds interpretability and accelerates translation while accumulating layered evidence for model robustness and portability across contexts.

Computational Evaluation

The performance validation of virtual cell models can be divided into two major aspects: computational evaluation and experimental validation^{9,133}. Computational evaluation primarily focuses on the model's fitting to existing data and its ability to predict unknown scenarios¹³³. Common quantitative indicators are as follows: (1) Reconstruction error and distribution consistency: This includes measuring the distance between the generated data and the training data in high-dimensional space, statistical distribution differences, and comparing their clustering structures through dimensionality reduction methods such as t-SNE⁵. (2) Prediction accuracy and stability: For extrapolated scenarios, the model's prediction is evaluated against real-world results^{134–136}. (3) Model complexity and generalization ability: This involves statistical model parameters, computational time, and evaluating the model's ability to adapt to new data through methods like cross-validation¹. Uncertainty quantification enhances the interpretability and robustness of conclusions. For example, Monte Carlo dropout can approximate the posterior predictive distribution to compute confidence intervals, and model ensembling can evaluate cross-model output consistency, thereby quantifying the reliability of virtual-cell model predictions^{137,138}. Driven by the destructive nature of single-cell assays—which precludes repeated measurements on the same cell¹³⁹—evaluation should pivot to the population-distribution level by comparing the statistical distributions of perturbed simulated cell populations with those observed experimentally⁹³. Accordingly, distributional distances—such as the Wasserstein distance, Kullback–Leibler divergence, and maximum mean discrepancy—serve as summary statistics to quantify agreement between model predictions and experimental data at the overall level. It is important to note that for high-dimensional biological data, “prediction accuracy” itself needs to be carefully defined and interpreted^{140–142}. Therefore, a multi-layered evaluation index system should be designed based on specific tasks to comprehensively quantify the performance of virtual cell models^{143–145}.

As virtual cell models become increasingly complex, the computational tools used to assess them are also evolving^{1,6,146}. Through multi-angle, multi-index computational evaluation, models with superior performance and robust stability can be initially screened for further experimental validation^{146–148}.

Experimental verification

The models, after computational evaluation, need to be validated through biological experiments to confirm whether their predictions reflect real biological phenomena^{9,149–153}. In terms of gene function prediction, CRISPR/Cas9 and other gene-editing technologies are used to verify the key targets predicted by the models through in vitro or in vivo validation^{154–156}. For example, this process is commonly used in the discovery of new targets to help confirm whether the candidate targets selected by the model are worthy of further development⁹.

New in vitro models comprise organoids and organ-on-chip systems. Organoids are three-dimensional cell aggregates grown in vitro that partially recapitulate the structure and function of native tissues; organ-on-chip platforms culture cells within microfluidic devices using microengineering techniques to emulate organ-level microenvironments and functions. These innovative platforms offer physiologically relevant testbeds to validate

Table 3 | Overview of virtual-cell/cell-level modeling platforms

Platform	URL	Strengths	Limitations	References
VCell	https://vcell.org	Unified environment for non-spatial/spatial, deterministic/stochastic, rule-based modeling; integrates ImageJ; batch simulation and sensitivity analysis; widely used in Ca ²⁺ signaling and receptor–ligand kinetics	Steep learning curve; high compute for complex 3D; configuration complexity for rule-based–spatial coupling	318 94 95
COPASI	https://copasi.org	ODE/SDE/Gillespie biochemical network modeling; parameter estimation, sensitivity, steady-state/time-domain analyses; SBML (Systems Biology Markup Language) support	Limited spatial modeling; multicellular interactions need external tools; visualization relatively limited	96 97 98 99
PhysiCell	https://physicell.org	Tissue-scale agent-based 3D modeling; diffusion fields, mechanics, immune microenvironment; PhysiCell Studio for visualization	Intracellular pathways need SBML/ODE coupling; limited native rule-based chemistry; HPC tuning requires experience	110 111 112 113
CompuCell3D	https://compuCell3d.org	Cellular Potts/GGH multiscale platform; reaction–diffusion and mechanics coupling; GUI + Python	Large intracellular networks require external coupling; very large models are time-consuming	114 115 116 117
Morpheus	https://morpheus.gitlab.io	Declarative XML; cell-based + ODE/PDE (partial differential equation) reaction–diffusion; good for rapid prototyping and reproducibility	Parallelism/custom numeric cores less flexible than code frameworks	118 119
Smoldyn	https://www.smoldyn.org	Particle-based spatial stochastic simulator; excluded volume, surface reactions; Python bindings	Best for small/medium systems; complex geometries need extra preparation	120 121
MCell/CellBlender	https://mcell.org	3D Monte Carlo reaction–diffusion; deep Blender geometry integration; BNGL (BioNetGen Markup Language) rules	High modeling threshold; heavy computation	102
BioNetGen	https://bionetgen.org	Canonical rule-based modeling; automatic network expansion; interoperable with PySB/MCell	BNGL has learning curve; spatial/multicellular needs coupling	100 101 102
PySB	https://pysb.org	Python DSL; supports BNG/Kappa; composable, versioned, testable models	Requires Python; spatial modeling needs external engines	103
OpenCOR/CellML	https://opencor.ws	CellML-based cross-platform simulation; rich Physiome ecosystem; strong for cardiac electrophysiology	Equation-centric; lacks direct multicellular/spatial microprocess simulation	125 126
CellNOpt	https://saezlab.github.io/CellNOptR/	Logic/fuzzy/differential logic signaling models fit to perturbation data	Limited support for continuous dynamics; needs high-quality perturbation datasets	104 105 106
CellCollective	https://cellcollective.org	Online collaborative logic-network modeling and large-scale simulation; easy model sharing	Lacks quantitative dynamics and spatial description	107
VCBA	—	In vitro tox and biokinetics virtual-cell platform; dose–time and organelle-level processes	General signaling networks weaker; high parameter demands	9 108 109
Chaste	https://chaste.github.io	High-performance C++; cardiac electrophysiology, cancer, soft tissue; rigorous testing	Complex build/config; molecular networks require coupling	124
ReaDDy	https://readdy.github.io	Particle-based reaction–diffusion with crowding and interaction potentials; Python API	Molecular/mesosopic focus; tissue-level phenomena require coupling	122 123
DeepCell	https://deepcell.org	Deep learning for cell image segmentation/phenotyping; cloud inference	Focused on imaging; limited support for dynamics simulation	128
CellPose	https://www.cellpose.org	Universal cell segmentation; adaptable to diverse modalities	Does not directly predict state changes	129 130 131
NVIDIA Modulus	https://developer.nvidia.com/modulus	PINN framework; supports biokinetic modeling	Requires strong GPUs; relies on prior equations	
OpenSim/Physiome	https://simtk.org/projects/opensim	Biomechanics and physiology; multiscale integration	Emphasizes mechanics; molecular pathways need external tools	125 127

predictions generated by virtual-cell models^{79,157}. Furthermore, in drug response prediction, the virtual cell model's predictions are validated through in vitro cell experiments or organoid models^{6,28,158}. For instance, if the model suggests that a particular candidate compound may induce hepatocyte steatosis or other toxic reactions, its toxicological indicators can be verified in mouse models^{158–160}. For the prediction of drug absorption, distribution, metabolism, and excretion properties, in vivo pharmacokinetic experiments can be conducted to validate the model's inference about absorption pathways^{6,28}. With the development of novel in vitro models, experimental systems that more closely mimic the in vivo environment are also being used to validate the predictions of virtual cells^{6,28,161}. These 3D models highly replicate the microenvironment of native tissues at multiple

levels, including cellular composition, tissue structure, and functional performance¹⁶¹. These models are expected to be deeply integrated with virtual cells to accelerate the transition of drugs from virtual screening to clinical trials^{6,161} (Fig. 3).

Virtual-cell–experiment convergence is validating drug-effect predictions. Several concrete studies show that predictions from virtual cell models closely match results from wet-lab experiments. Driven by integrated modules for drug exposure, cardiomyocyte–drug interaction, and system-level response, Wang and colleagues built an in-vitro–in-vivo translation platform based on human iPSC (induced pluripotent stem cell)-derived cardiomyocytes (hiPSC-CMs). Its simulations predicted the incidence of cardiotoxicity for the anthracycline doxorubicin and the anti-HER2

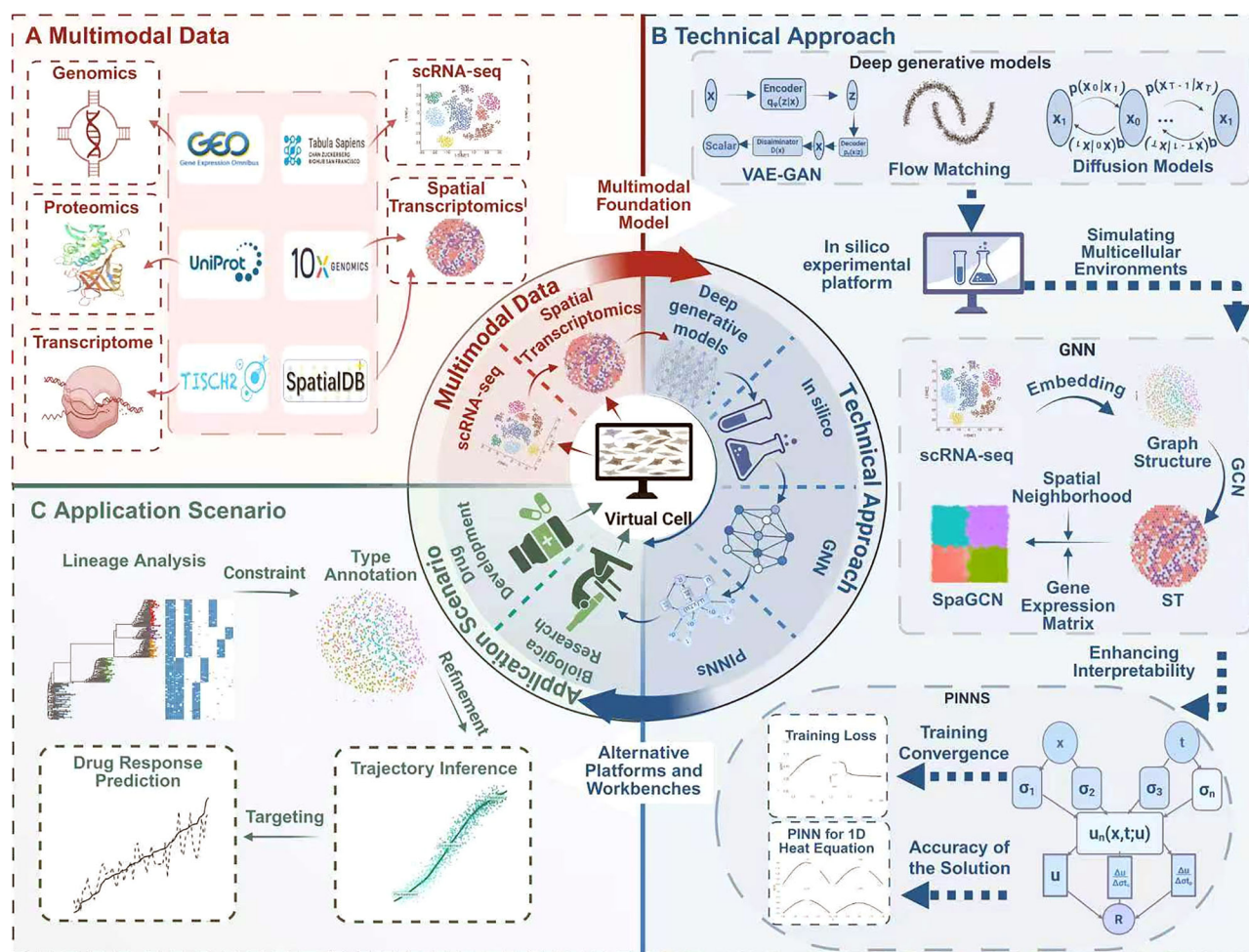


Fig. 2 | Schematic workflow of virtual cell models. **A** Multimodal data and representative resources include genomics, proteomics, and transcriptomics data types with mainstream databases and platforms, with a Multimodal Foundation Model serving as the hub that connects data to methods; **B** The technical routes comprise three paths: Deep generative models, Graph neural networks, and Physics-informed neural networks, where Deep generative models encompass VAE-GAN, Flow Matching, and Diffusion Models and connect to an in vivo/in vitro-referenced in silico experimental platform, Graph neural networks center on joint graph construction from single-cell and spatial transcriptomics and use SpaGCN as a

representative implementation, and Physics-informed neural networks target Training Convergence and Accuracy of the Solution and present a PINN for the 1D Heat Equation to strengthen interpretability and physical consistency; **C** Application scenarios cover lineage analysis and cell-type annotation with constraint information to improve labeling robustness and include drug response prediction and trajectory inference with Targeting and model Refinement cues that form a feedback loop from applications to methods and data, while Alternative Platforms and Workbenches at the lower rim indicate where diverse tasks are executed and validated computationally. (Created in <https://BioRender.com>).

antibody trastuzumab with high concordance to clinical observations¹⁶². Feasibility is demonstrated: hiPSC-CM-based virtual cells enable animal-free, quantitative prediction of anticancer-drug-induced cardiac dysfunction. Similarly, the Virtual Cell-Based Assay (VCBA) has been used to simulate intracellular and organellar pharmacokinetics. Driven by mechanistic transport and binding representations, Worth et al. used VCBA to model the distribution of selected compounds in HepaRG hepatocytes and cardiomyocytes and their effects on mitochondrial membrane potential. The simulations accurately recapitulated in-vitro changes in toxicity readouts for the uncoupler FCCP, caffeine, and amiodarone⁹. VCBA thus maps intracellular and mitochondrial concentrations to membrane potential shifts and cytotoxicity, reinforcing the agreement between virtual predictions and in vitro data.

“Next-generation in-vitro models” include organoids and organs-on-chips: organoids are three-dimensional cell aggregates cultured ex vivo that partially reproduce tissue architecture and function, whereas organs-on-chips culture cells in microfluidic devices to emulate organ microenvironments and functions^{79,157}. These systems provide physiologically relevant platforms for validating virtual-cell predictions. Driven by patient-specific

epithelium-immune crosstalk, patient-derived intestinal organoids co-cultured with immune cells have been used to assess on- and off-tumor toxicities of T-cell bispecific antibodies. Harter et al. showed that an EpCAM-targeted T-cell-engaging bispecific antibody induced apoptosis in healthy intestinal organoids, aligning with toxicities reported clinically¹⁶³. Result: faithful reproduction of target-mediated toxicity in complex microenvironments, providing a reliable experimental anchor for validating virtual predictions. Driven by a policy shift toward animal reduction, the FDA (Food and Drug Administration) has recently signaled broader use of organs-on-chips, virtual models, and organoids in preclinical safety assessment¹⁶⁴.

Collectively, these exemplars substantiate the effectiveness of virtual-cell models for predicting drug toxicity and ADME behavior. Driven by cross-validation against hiPSC-derived cardiomyocytes, organoids, and animal studies, virtual-cell platforms are accelerating drug development and enabling human-physiology-aligned alternatives to traditional animal testing. Regulatory agencies, including the FDA, are encouraging adoption of these integrated computational-biological approaches, steering screening and safety evaluation toward human relevance.

Model Validation Mechanism

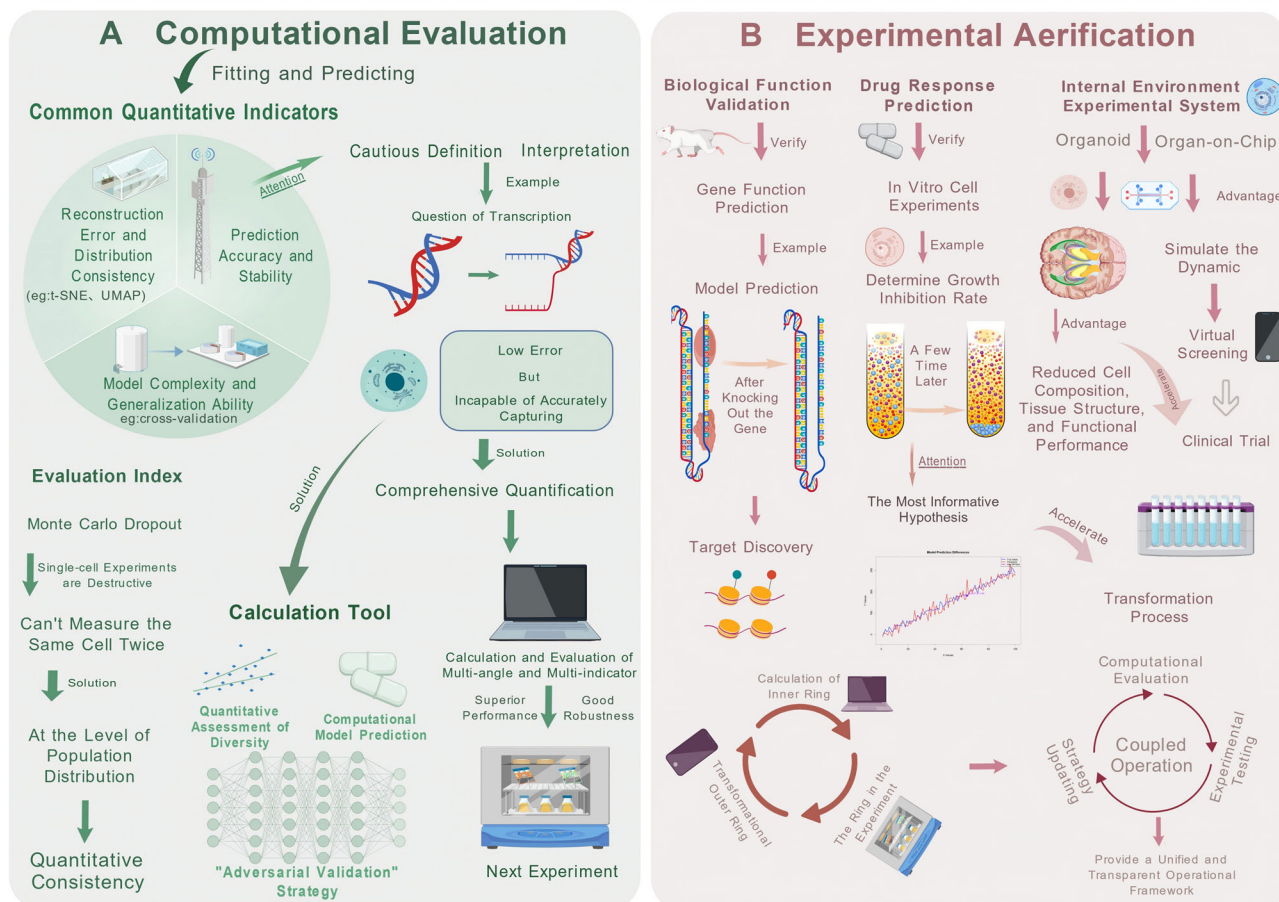


Fig. 3 | Schematic diagram of virtual cell model verification mechanism.

A Computational evaluation: Quantitative indicators of distribution consistency and predictive ability. The figure shows commonly used quantitative indicators for computational evaluation and the computational tools to assess them. Common quantitative indicators include reconstruction error and distribution consistency, prediction accuracy and stability, model complexity and generalization ability. The computational tools to evaluate them include those based on cell phenotype space coverage, those evaluated using Spearman's rank correlation coefficient, and those based on the "adversarial validation" strategy. Through multi-angle and multi-index computational evaluation, models with better performance and good robustness can be initially selected for the next stage of experimental verification. **B Experimental**

verification: Process from CRISPR target editing to drug response testing. The figure shows three experimental verification methods, including biological experimental verification, drug response prediction, and in vivo environmental experimental systems with the development of new in vitro models. Biological experiments are used in gene function prediction by knocking out or overexpressing a certain gene for observation to discover new targets; drug verification validates the predictions of virtual cell models through in vitro cell experiments or organoid models; in vivo environmental experimental systems include organoids, organ-on-a-chip, etc. With the standardization and wide application of verification models, these models can be deeply integrated with virtual cells to accelerate the transformation process of drugs from virtual screening to clinical trials. (Created in <https://biogdp.com>).

Closed-loop integration of computational evaluation and experimental validation

Closed-loop architecture—comprising a computational inner loop, an experimental middle loop, and a translational outer loop—links methodological assessment with application-level verification; entry, decision, and exit gates at key nodes operationalize bottom-up continuous improvement. In the computational inner loop, strict data partitioning, batch harmonization, and pre-analysis quality control are prerequisites. We quantify distributional concordance under perturbation scenarios and characterize predictive uncertainty, with primary criteria being a pre-registered concordance threshold and an upper bound on the confidence interval width. If criteria are not met, the workflow reverts to data cleaning, structural-prior specification, and regularization tuning¹⁶⁵. Driven by near-physiological systems (e.g., organoids and organ-on-chip platforms), the experimental middle loop subjects model claims to targeted tests under pre-specified functional endpoints and statistical power. When effect direction or magnitude diverges from computational expectations—or power is insufficient to reject the null—the loop feeds back to joint revisions of modeling

assumptions, feature selection, and experimental-protocol parameters¹⁶⁶. In the translational outer loop, cross-platform replication and prospective datasets constitute the primary settings. Deployment readiness is adjudicated by integrated evidence on generalizability, stability, and interpretability, supplemented by safety review and failure-mode analysis; if out-of-domain performance departs from predefined bands, feedback triggers coordinated optimization of feature engineering and experimental design¹⁶⁷. Tri-loop coupling secures distribution-level statistical reproducibility with bounded uncertainty while cross-validating functional evidence and translational feasibility, yielding a unified and transparent operating framework for comparable evaluation and reproducible validation across applications.

Application Scenarios

Post-validation priority: establishing the model's real-world and translational value. Applications of virtual cells are steadily expanding to drug screening, mechanistic inference, digital twins, and multi-platform interoperability, opening new avenues for precision medicine.

Virtual-cell-enabled precise screening and mechanistic inference

The virtual cell model can assist in advancing and refining drug screening processes^{1,5,6,9,10}. On one hand, during the target discovery phase, virtual cells can simulate gene knockout or overexpression, predicting downstream molecular networks and phenotypic changes to infer potential therapeutic targets and their mechanisms of action^{1,9}. Researchers can prioritize the validation of targets predicted by the model to have significant effects, improving the hit rate of experimental screening¹⁶⁸. On the other hand, in lead compound screening, virtual cell models can be used for in silico screening of large numbers of candidate compounds, predicting their ability to correct pathological cell states and assessing their risk of toxic side effects^{6,10}. Regarding mechanism inference, virtual cell models can provide in-depth analysis of the dynamic processes of drug-cell interactions^{5,6,10}. Moreover, virtual cells can also be used for hypothesis testing. Researchers can introduce a new regulatory factor or pathway hypothesis into the model to test whether it can reproduce the observed phenomena⁵. If the model validation supports the hypothesis, targeted experimental validation can then be designed at the experimental level^{169–171}. This approach is expected to improve the efficiency of mechanistic research, quickly identifying key factors and pathways with the support of virtual cells^{169–171}.

It is important to emphasize that the use of virtual cells should always be closely integrated with biological experiments^{172–174}. In the short term, virtual results serve more as supporting evidence^{175,176}, thus hypotheses generated through model predictions must be experimentally validated, with feedback used to refine the model and continually improve its consistency with reality¹. This closed-loop process will drive the true precision of drug screening and mechanism analysis, enabling the efficient translation of vast amounts of data into knowledge discovery^{177,178}. An integrative evaluation workflow, distilled from a systematic synthesis of prevailing standards and practices, enables comparable assessment and reproducible validation of virtual cell models across application contexts. We benchmark this workflow against existing methodological frameworks, specifying shared steps—data standardization, identification of bias sources, distribution-level concordance testing, and uncertainty quantification—and modules unique to virtual-cell settings—distributional distance evaluation for perturbation simulations, cross-modal alignment, and mapping of functional endpoints between organoids and organ-on-chip platforms. Driven by cell-state prediction and perturbation-response inference as core tasks, the workflow applies to studies with traceable data provenance and cross-batch calibration of experimental systems. Limitations: cross-species generalization and extrapolation to ultra-rare cell populations warrant particular caution.

Synergy between digital twins and virtual cells

The introduction of the concept of “Digital Twin” has provided new opportunities for the application of virtual cell models⁶. A digital twin typically refers to a high-fidelity virtual replica of a specific entity, which can receive real-time data from the physical entity and perform simulations^{179–181}. By integrating virtual cell models into the digital twin framework, a multi-scale, comprehensive simulation from the molecular and cellular level to the whole organism can be achieved^{1,6,10,161,182,183}. In rare disease research, by combining patient-specific stem cell-derived virtual models with patient phenotype data, the effects of personalized drug interventions can be simulated^{161,182}. This strategy allows for large-scale virtual screening of efficient, low-toxicity candidate compounds before clinical trials¹⁰ and has shown potential to partially replace animal experiments in target discovery and other processes^{6,9}.

The combination of digital twins and virtual cells also provides a new approach to disease prediction and monitoring^{6,181}. By continuously inputting biomarker data from patients, virtual cell models can simulate the cellular evolution of disease progression, offering doctors predictions regarding the disease’s course and outcome⁶. For instance, in cancer treatment, if the virtual model predicts changes related to drug resistance signals, it can provide an early warning and suggest treatment adjustments^{184–188}.

These applications rely on high-quality individual data and the model’s ability to sensitively capture individual differences^{187,189–191}. With advancements in data collection and modeling techniques, the integration of digital twins and virtual cells will become more seamless, playing a more active role in clinical decision support^{187,188,192}.

Boundary setting and complementarity

Despite the powerful capabilities of virtual cell models, they cannot fully replace traditional methods in all scenarios. Therefore, it is essential to define their application boundaries and leverage their complementary role⁶. In the context of complex multicellular behaviors, virtual cells currently focus on molecular and intracellular processes. For phenomena involving tissue morphological changes, traditional multicellular modeling or experimental approaches remain indispensable¹. Regarding model interpretability, physical models and rule-based models, which are directly grounded in known human laws, are generally more readily accepted by researchers. In contrast, although deep learning-driven virtual cells provide accurate predictions, they may struggle to offer deductive mechanistic explanations¹⁶. Thus, in studies requiring a clear mechanistic understanding, virtual cells serve as an exploratory tool, while the final mechanistic elucidation should be combined with traditional experiments and analytical models¹. Additionally, virtual cells complement other emerging technologies^{5,6,193}. For instance, when combined with single-cell lineage tracing technology, they can provide developmental pathways, which are difficult for models to simulate, serving as ground truth to calibrate model parameters⁵. Researchers should fully recognize the strengths and limitations of virtual cells and integrate them with other practice methods: capitalizing on their speed and scalability while using traditional methods to ensure reliable results and clear explanations^{194–197}. Only in this way can virtual cell models find their optimal positioning and achieve maximum benefits in biomedical research and drug development^{198–200}.

Virtual cells and cell-level modeling platforms: status and comparative analysis

Mechanism-driven alignment with the target scenario is the primary determinant of successful application. The selection of a modeling platform is driven by (i) the biological endpoints of the question and (ii) the complexity of implementing the pertinent molecular mechanisms. When the aim is Section 4.1’s precision screening and mechanistic deduction, intracellular network platforms are indicated—e.g., COPASI^{96,97} for simulating drug impacts on pathway behavior, or CellNOpt^{104,105} for data-driven reconstruction of signaling circuitry under perturbations. For Section 4.2’s digital-twin use cases involving multi-cellular coordination, tissue-level simulators—PhysiCell^{110,111} or CompuCell3D^{114,115}—are preferable to capture tumor expansion and integrated tissue responses to drugs. Readers may consult Section 2.6’s computational-speed overview and Table 1’s comprehensive comparison (Fig. 4) to select the most suitable platform.

Clinical translation, ethics, and compliance

Driven by collective socio-technical shaping, the safe translation of virtual cell models from theoretical constructs to clinical use has become the decisive inflection point. The trajectory is conditioned not only by technical capability but by regulatory architectures, ethical safeguards, legal accountability, and trust-building mechanisms. We outline how compliance regimes, data privacy security, intellectual property ownership and stewardship, and fairness auditing collectively facilitate clinically acceptable deployment, paving the way for the field’s next phase.

Regulatory Trends

With the rapid development of non-animal testing methods, AI-driven virtual cell models have gradually entered the discussion within regulatory agencies^{6,16,201,202}. In 2022, the U.S. Congress passed the FDA Modernization Act 2.0, which explicitly stated that animal testing data would no longer be a mandatory requirement for new drug IND submissions. It allowed for the use of humanized cell models, organoids, organ-on-a-chip systems, and

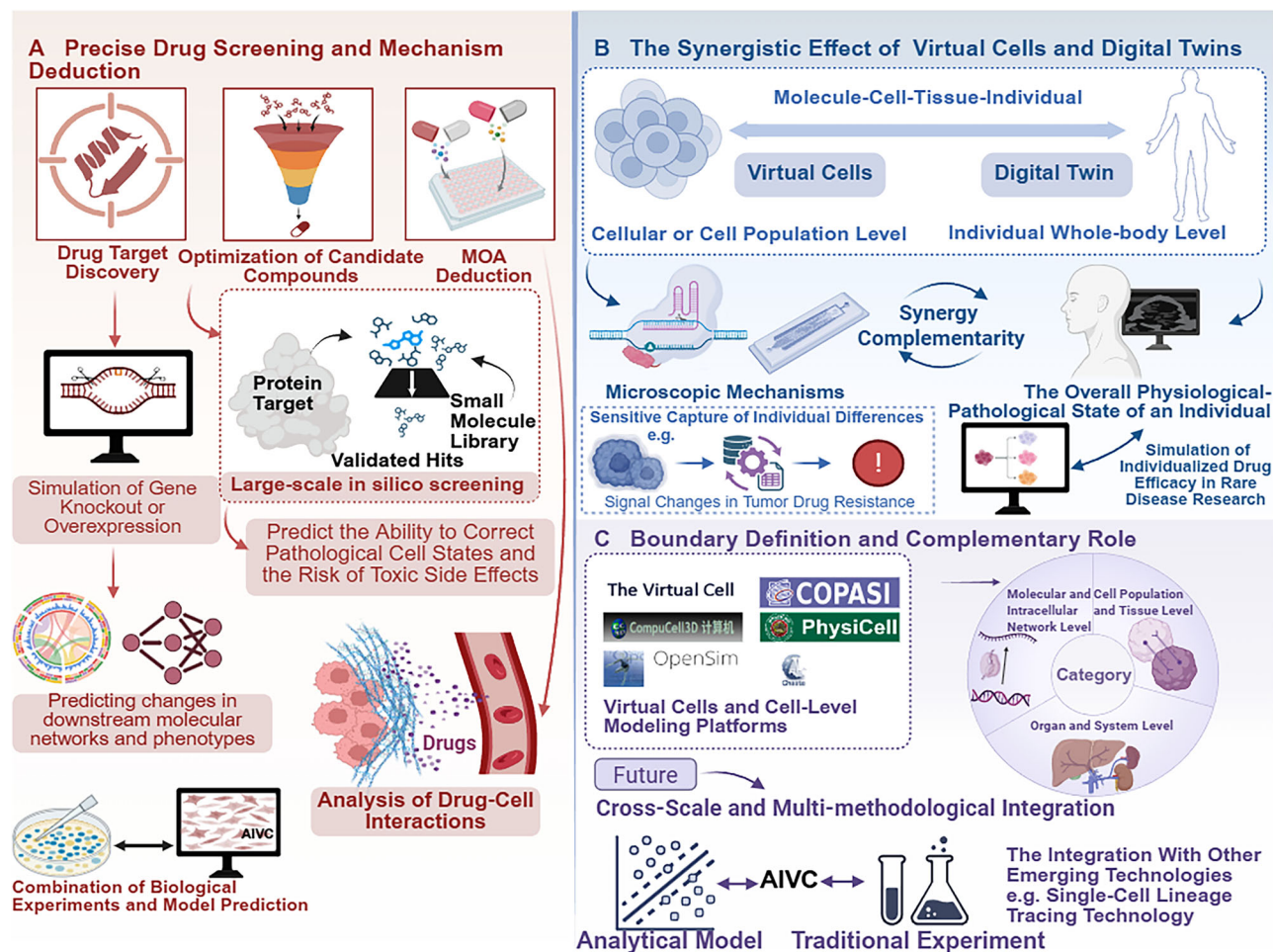


Fig. 4 | Applications of AI-Driven Virtual Cell Models in Preclinical Research. This figure elucidates the diverse applications of AI-driven virtual cell models in preclinical research, encompassing three dimensions: precision drug screening & mechanism deduction, synergistic integration with digital twins, and boundary definition, complementary roles, along with platform classification. **A** Precision Drug Screening & Mechanism Deduction It demonstrates the core value of virtual cells in drug development, spanning drug target discovery, optimization of candidate compounds, and model predictions enables precise drug screening and mechanistic interpretation. **B** Synergistic Integration with Digital Twins This section illustrates their complementary value in multi-scale simulation: virtual cells focus on microscopic mechanisms at the cellular or cell population level (e.g., sensitive capture of individual differences in tumor drug resistance signals), while digital twins encompass the overall physiological-pathological state of an individual. Their synergy achieves multi-scale integration from molecule-cell-tissue to individual,

exemplified by personalized drug efficacy simulation in rare disease research. **C** Boundary Definition, Complementary Roles, and Cell-Level Platform Classification. Boundary and Complementarity: Despite their capabilities, virtual cell models cannot fully replace traditional methods across all scenarios. They excel in molecular and intracellular processes but rely on traditional experiments or analytical models to explain complex tissue morphogenesis or provide deductive mechanistic. Meanwhile, they complement emerging technologies such as single-cell lineage tracing. Platform Classification: Cell modeling platforms are categorized by scale: the molecular and intracellular network level (e.g., COPASI for biochemical network dynamics), the multicellular-tissue level (e.g., PhysiCell for tissue-scale agent-based modeling, CompuCell3D for Cellular Potts-based simulation), and the organ-system level (e.g., OpenSim for biomechanical modeling). Future development trends lean toward cross-scale and multi-methodological integration. (Created in <https://BioRender.com>).

artificial intelligence/machine learning (AI/ML) models as alternatives^{6,203,204}. This legislation opened a potential path for the regulatory and compliant application of virtual cells, particularly in early-stage toxicology and pharmacology evaluations^{6,9,205}. In 2025, the U.S. FDA released its first draft guidance on the use of AI models in drug development, titled “Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drugs and Biologics,” proposing a risk-based framework for model credibility assessment²⁰⁶. In Europe, the European Chemicals Agency (ECHA) and the EU REACH (Registration, Evaluation, Authorization and Restriction of Chemicals) regulations have also strengthened the adoption of non-animal testing methods in recent years, with some toxicology evaluation frameworks incorporating computational modeling and bioinformatics prediction tools^{207,208}. Overall, global regulatory trends are cautiously supportive of AI-driven alternative methods: on one hand, they recognize their potential in accelerating drug

development and reducing animal use, while on the other hand, they ensure their reliability and transparency through legislation and guidelines when used in decision-making^{6,16,203,204}. For virtual cell model developers, understanding and adhering to these policy guidelines and considering regulatory requirements during the model design phase, will help increase the likelihood of success in subsequent clinical translation submissions^{209,210}.

Regulatory and Implementation Challenges

Despite the gradual relaxation of the policy environment, virtual cell models still face significant challenges in fully replacing traditional *in vivo* and *in vitro* experiments under the current regulatory system^{6,12,182}. The U.S. FDA still requires the submission of adequate *in vitro* and/or *in vivo* validation data during the IND review process, with AI models currently considered as supplementary evidence rather than the sole basis⁶. Differences in data sources, model architecture, parameter settings, and validation

processes among research teams, along with the lack of a unified third-party benchmark testing platform, limit the comparability and reproducibility of model results^{16,211}. At present, regulatory agencies have not established clear review guidelines for AI-based biological models, and projects are often subject to case-by-case review, making it difficult for applicants to predict the required level of validation^{16,211,212}. For example, there is no unified standard regarding the level of predictive accuracy required to support an IND application, and regulatory demands may vary significantly across different cases⁶. Moreover, regulatory agencies' personnel must also address challenges^{213–215}. Reviewing a complex virtual cell model typically requires interdisciplinary experts in computational biology, machine learning, and disease biology. However, many regulatory agencies currently lack sufficient human resources in this area, potentially leading to delays or uncertainty in the review process^{16,211}. As a result, agencies like the FDA are improving AI review capabilities through internal training and external consultants, but this cognitive gap remains a significant barrier to model translation in the short term^{216–223}.

Another challenge arises from data sharing and privacy issues^{224–231}. High-quality virtual cell models typically require the integration of patient-derived data for training, which often involves sensitive personal health information^{226,232–234}. When submitting regulatory reviews, applicants need to provide sufficient information to demonstrate the reliability of the model while safeguarding patient privacy. This has led to the development of innovative solutions, such as secure multi-party computation and federated learning, which enable multiple institutions to collaborate on model validation without sharing raw data^{232,235,236}. However, these technologies also add complexity and uncertainty to the review process^{237–239}. Furthermore, legal issues such as model intellectual property ownership and compliance with training data regulations need to be clarified in submission materials, as failure to do so may trigger compliance investigations and delay the review process⁶. Finally, the application of AI models in regulatory decision-making lacks successful precedents²⁴⁰. To date, there are few cases where new drugs have been approved based solely on AI model results, with AI still predominantly used as an auxiliary tool^{241,242}. At the regulatory implementation level, virtual cell models continue to face challenges related to model standards, review capabilities, data compliance, and successful demonstrations, requiring joint efforts from industry and regulators to resolve^{2,182,243}.

Cross-scale coupling still constrains the transition from single-cell models to tissue-level applications. At the single-cell level, virtual models must be embedded within higher-order physiological contexts, encompassing cell–cell interactions, three-dimensional tissue architecture, and whole-body dynamics²⁴⁴. Driven by multiscale integration, existing frameworks (OpenSim, Physiome) can host cell-level models at organ and organism scales^{127,245}, yet they prioritize macroscopic processes and remain limited in their explicit representation of signaling pathways and gene-regulatory mechanisms. Consequently, after *in vitro* verification, virtual cells require intermediate testing and parameter calibration in organoids and animal models to progressively approximate real-world patient contexts^{246,247}. Clinical translation hinges on parallel progress in cross-scale technical concordance and regulatory–ethical alignment, both of which should be treated as coequal bottlenecks for virtual cells.

Data privacy and security compliance

Virtual cell models are constructed and trained on large-scale biomedical datasets that often contain sensitive patient information^{7,8}. If mishandled, such data can lead to breaches of personal privacy and raise ethical concerns^{5,226,233,234,248}. Many countries and regions have enacted laws governing the use of health data; for example, China's Personal Information Protection Law (PIPL) requires health data to be stored locally and mandates obtaining the data subject's explicit consent²⁴⁸, while the United States' Health Insurance Portability and Accountability Act (HIPAA) requires encryption and de-identification of protected health information (PHI)^{6,248}. European context: mandatory GDPR (generative adversarial network) -compliant protection of patient data privacy²⁴⁹. In parallel, stakeholders

should closely follow the forthcoming European Health Data Space (EHDS) initiative, which aims to enable secure sharing and standardized reuse of research data within a defined regulatory framework²⁵⁰. These regulations mean that developers of virtual cell models must ensure legal compliance when collecting and processing data, and that cross-border data transfers may give rise to legal conflicts that affect the use of models in international multicenter studies^{248,251}. Beyond regulatory compliance, technical measures are also needed to safeguard data security and privacy^{252–259}. Evidence indicates that deep generative models can “memorize” unique features of training samples and, in extreme cases, reproduce portions of the original training data in their outputs¹⁶. To mitigate such risks, developers should deploy privacy-preserving mechanisms when releasing and sharing models^{260–266}. For example, differential privacy techniques can inject calibrated noise into model parameters to obscure individual-level features and thereby reduce potential privacy leakage with minimal impact on model performance^{16,267–269}. Likewise, federated learning frameworks allow models to be trained on local institutional data while only sharing model weight updates, thereby architecturally avoiding the risk of centralized data exposure^{232,270,271}. A combined legal and technical approach is therefore required to realize the value of virtual cell models while ensuring data security^{272,273}. Cybersecurity is another essential consideration. If virtual cell models are provided to healthcare institutions as cloud services, the databases and computing infrastructure on which they depend may become targets for attack^{274–276}. Consequently, model deployment should be accompanied by strict access controls, vulnerability scanning, and incident-response mechanisms, and model outputs should be subject to anomaly monitoring so that any aberrant behavior triggers timely alerts and intervention^{277–279}. These measures will help ensure that virtual cell models, as digital medical products, meet security standards at least as stringent as those applied to conventional clinical medical software^{6,280}.

Intellectual property and liability

The ownership of intellectual property in outputs generated by AI models currently lacks a unified legal definition^{6,16}. In the commercialization of virtual cell models, the boundaries of rights and interests among data providers, model developers, and end users remain unclear. Moreover, as a novel digital product, the regulatory and legal status of virtual cell models is still being explored⁶. When model predictions are used to support critical R&D or clinical decision-making, and an erroneous prediction leads to adverse outcomes, determining which party should bear responsibility—as well as disputes over the allocation of commercial benefits—becomes contentious^{16,281}. For example, if a model is trained on large volumes of patient single-cell omics data provided by a hospital, should the intellectual property rights to a drug target predicted by the model belong to the model company or to the data-providing hospital^{282–284}? Similarly, if the algorithm is developed by an AI company but a pharmaceutical firm uses the model to discover a new molecule, how should the related intellectual property be shared^{282–284}? These questions currently have no precedent and must be progressively clarified in practice through contracts and legislation to address real-world problems^{6,16}. Model developers, providers, and users may shift liability among themselves, which impedes compensation for harmed parties and undermines healthy industry development.

Therefore, regulatory authorities and the legal community should intervene early to establish liability-determination frameworks and insurance mechanisms tailored to AI model applications, clarify the rights and responsibilities of all parties, and reduce the legal risks associated with adopting new technologies^{285–287}. Overall, clarifying intellectual property and liability allocation is a necessary condition for translating virtual cell technology from the laboratory into industry, requiring coordinated advancement through both legislation and industry self-regulation^{285–287}.

Algorithmic fairness and explainability

The fairness and transparency of virtual cell models directly influence their adoption in clinical settings^{16,133}. If the training data contain biases related to race or other factors, models can amplify these injustices and systematically

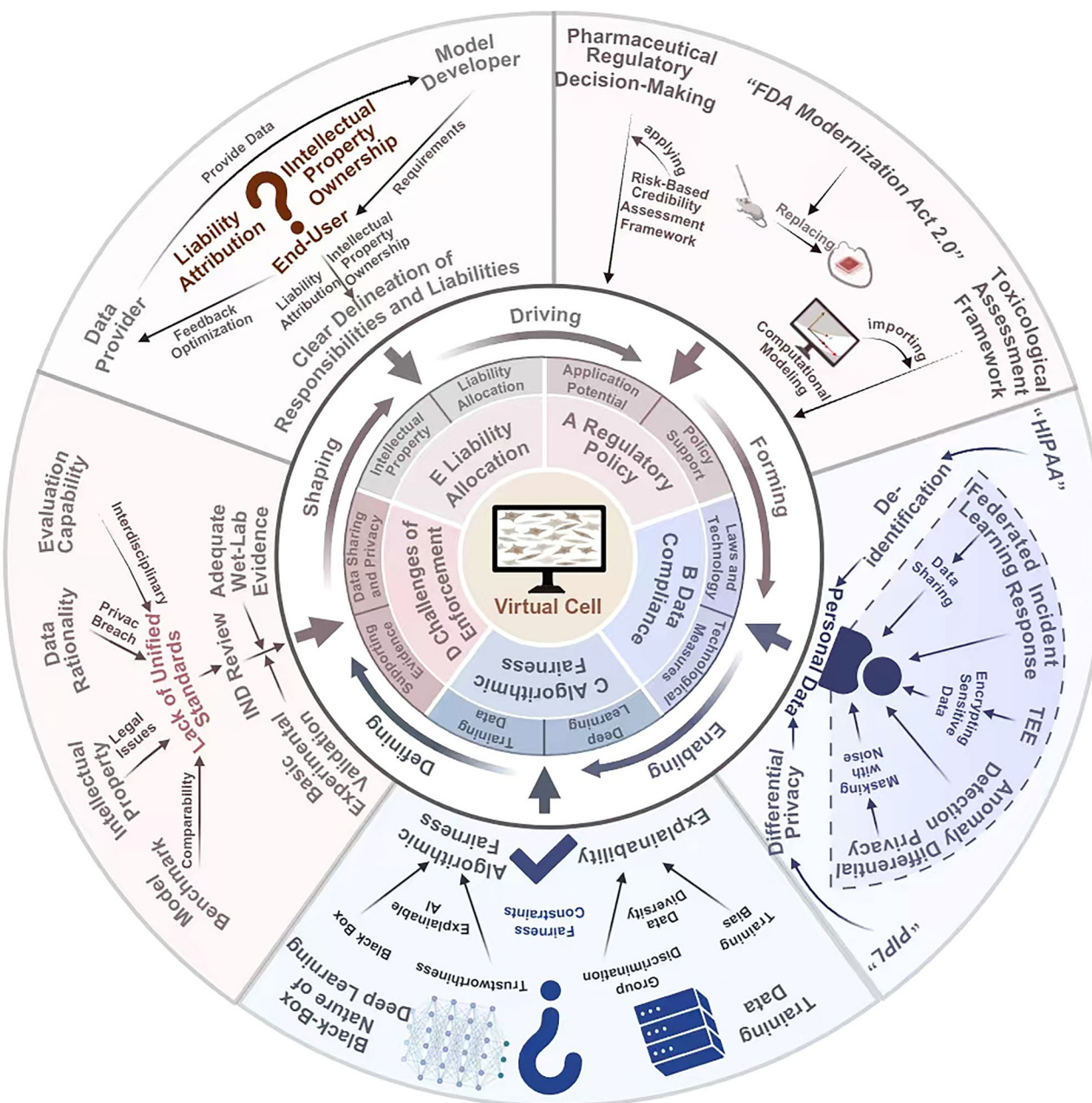


Fig. 5 | Clinical translation, ethics, and compliance challenges of virtual cell models. Virtual Cell governance and compliance translational pathway. **A** The regulatory-policy core locates Virtual Cell within risk stratification, credibility assessment, and pertinent provisions, and specifies its interface with drug-regulatory decision-making; **B** The data-compliance core covers requirements for data acquisition and cross-domain transfer, together with technical pathways such as de-identification, federated learning, and trusted execution environments; **C** The algorithmic-fairness core emphasizes interpretability calibration and bias detection

and constraint at both population and individual levels, culminating in a model credibility report; **D** The application-potential layer delineates the use boundaries and evidence needs of Virtual Cell in drug evaluation, toxicological inference, and scenario-based decision-making; **E** The responsibility-allocation layer depicts the duties and rights of data providers, model developers, and end users regarding IP attribution, responsibility determination, and feedback-loop optimization. (Created in <https://BioRender.com>).

disadvantage certain groups in prediction outcomes^{1,16}. For example, if training data are drawn predominantly from Western populations, a model may perform with lower accuracy when predicting drug responses in East Asian populations, thereby reducing those groups' opportunities to receive effective therapies^{288–291}. Such AI bias should be mitigated by increasing data diversity and by incorporating fairness constraints during model training^{16,292}. Moreover, the “black-box” nature of deep learning models undermines researchers' and clinicians' trust in their predictions^{16,133}. Therefore, integrating explainable artificial intelligence (XAI) approaches into virtual cell models is essential^{1292–294}. Common XAI techniques—such as

SHAP (SHapley Additive exPlanations) value analysis and integrated gradients—can quantify the dependence of model outputs on individual input features, thereby illuminating the rationale behind model decisions^{292,294}. Through these methods, we can answer critical questions like “Why does the model consider a particular gene mutation pathogenic?” or “On what basis does the model predict that drug A is more effective than drug B?”²⁹². Introducing fairness constraints during model development and augmenting interpretability analyses during deployment will markedly enhance the credibility of virtual cell models and increase their likelihood of broad adoption in both research and clinical practice^{16,292} (Fig. 5).

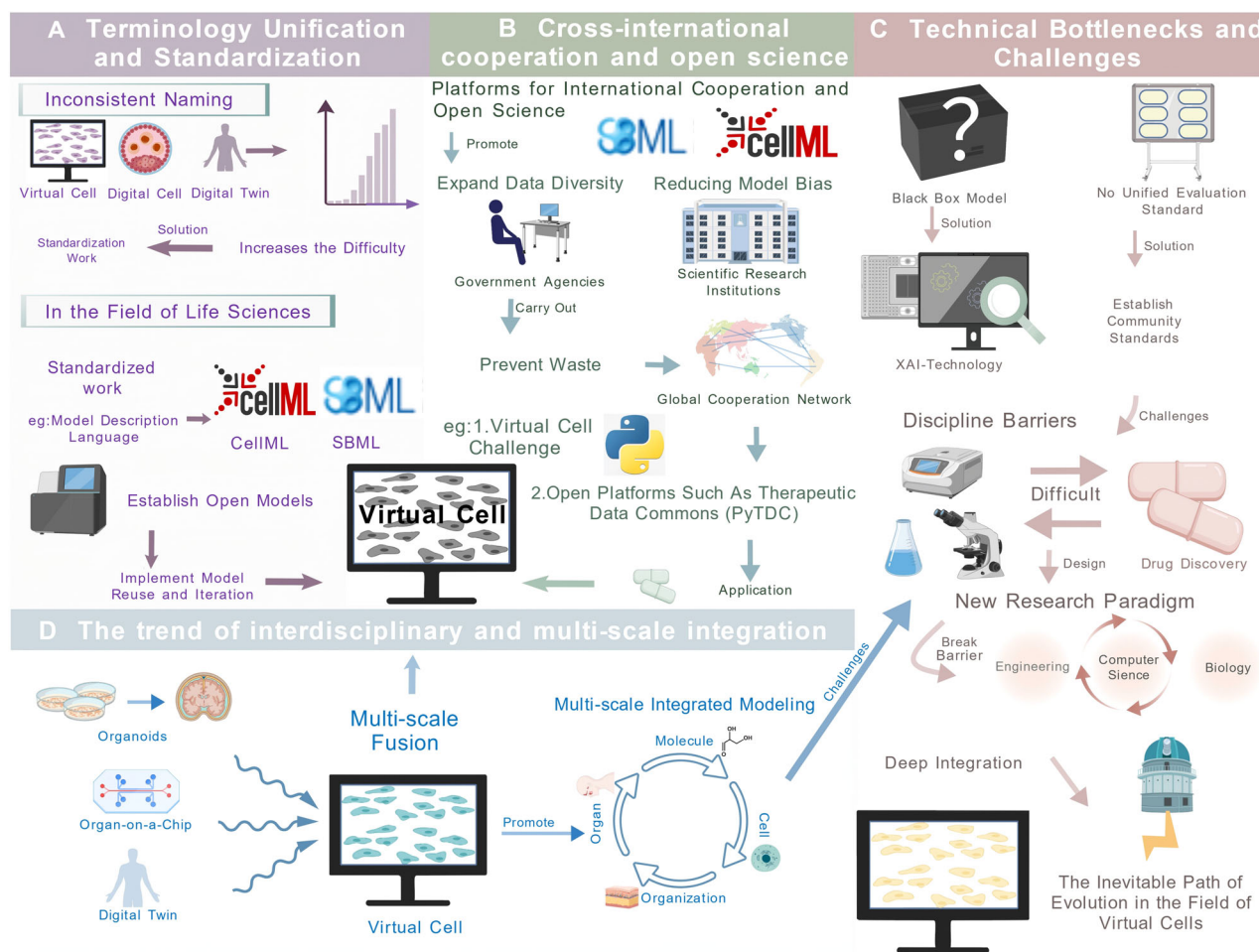


Fig. 6 | Schematic diagram of future development and prospects. **A** Terminology unification and standardization: The figure shows that terms such as “Virtual Cell”, “Digital Cell”, and “Digital Twin” are not used consistently in academia and industry. Therefore, the International Organization for Standardization (ISO) has begun to promote the unification of terms and definitions in the field of digital twins, and promote the use of standard model description languages (such as SBML, CellML) to describe virtual cell models. **B** International cooperation and open science: The figure shows that internationally, to accelerate the iteration and clinical transformation of virtual cell technology, it is necessary to establish platforms for international cooperation and open science, and promote unified model performance evaluation standards and regulatory adoption paths. Governments and scientific research institutions around the world are paying increasing attention to open science, and the virtual cell field is expected to emerge a global cooperation network

to promote the faster application of technology. **C** Technical bottlenecks and challenges: The figure shows black-box models, poor cross-platform compatibility, and lack of unified evaluation standards, and shows solutions. **D** Trend of interdisciplinary and multi-scale integration: The figure shows that virtual cells will be deeply integrated with cutting-edge technologies such as organoids, organ-on-a-chip, and digital twins to achieve multi-scale integrated modeling from molecule-cell-tissue-organ-individual. The challenge is to be limited to a single level or a single field, and it is necessary to explore new cooperation mechanisms and research paradigms to break down the barriers between biology, computer science, and engineering. In-depth cross-field integration and multi-scale data + model integration will become an inevitable path for the evolution of the virtual cell field. (Created in <https://biogdp.com>).

Future development and outlook

With regulatory–ethical–compliance alignment achieved, virtual-cell development enters a new phase: foundation in terminology harmonization/standardization; delineation of technical hurdles; prospect of cross-disciplinary, multiscale integration; and establishment of open-science, international-collaboration platforms to expedite innovation.

Terminology harmonization and standardization

At present, terms such as “Virtual Cell”, “Digital Cell”, and “Digital Twin” are not used consistently across academia and industry^{1,182}. This terminological overlap not only complicates cross-disciplinary communication but can also create ambiguities in regulation and review processes⁶. The International Organization for Standardization (ISO) has already begun promoting the harmonization of terms and definitions in the digital twin domain; however, in the life sciences, there is an urgent need for multinational academic organizations and regulatory authorities to take the lead

in developing a standardized vocabulary and classification framework for virtual cells⁶. Standardization efforts should extend beyond nomenclature to include model description languages, data formats, and validation metrics²⁹⁵. For example, promoting the adoption of standard model description languages (e.g., SBML, CellML) to represent virtual cell models would facilitate model sharing and reuse across different platforms^{125,296}; likewise, establishing public model and data repositories and encouraging researchers to submit validated models would enable model reuse and iterative improvement⁶. By unifying terminology and standards, the virtual cell research ecosystem will become more standardized, and collaboration among researchers will proceed more smoothly²⁹⁷.

Technical bottlenecks and challenges

The limited interpretability of virtual cell models represents a major bottleneck to their regulatory and clinical adoption¹⁶. As black-box models, they often fail to provide explicit biological causal explanations, undermining

clinicians' and regulatory experts' trust in their predictions¹. These issues are particularly pronounced in the preclinical stages of drug development, where the accuracy and reproducibility of model outputs directly affect the design and risk assessment of subsequent clinical trials²⁵¹. Developing a graded framework for model credibility is therefore a promising avenue, stratifying predictive confidence according to factors such as the volume of training data and the rigor of validation, and aligning each credibility tier with the verification intensity required for different application scenarios¹. Only by addressing these bottlenecks in a targeted manner can virtual cell models gain a foothold in more stringent regulatory and clinical environments⁶.

Cross-disciplinary, multiscale integration

Looking ahead, virtual cells will be deeply integrated with cutting-edge technologies such as organoids, organ-on-chip platforms, and digital twins, enabling multiscale, unified modeling that spans molecules, cells, tissues, organs, and whole organisms^{1,6,204}. Such multiscale integration is expected to more systematically predict drug responses and disease progression, thereby enhancing the role of virtual cells in precision medicine^{1,298}. At the same time, emerging generative artificial intelligence and large multimodal models will furnish virtual cells with stronger generalization and inferential capabilities^{299,300}. However, realizing the interdisciplinary, multiscale integration described above requires overcoming the current fragmentation of research: the vast majority of studies remain confined to a single level or domain, and disciplinary barriers impede model interoperability¹. Deep cross-domain integration and the consolidation of multiscale data modalities and model types are likely to become essential pathways for the evolution of the virtual cell field³⁰¹.

International collaboration and open science

To accelerate the iterative development and clinical translation of virtual-cell technologies, international collaborative and open-science platforms should be established^{1,302}. Such platforms could enable the sharing of large-scale multimodal omics datasets, the adoption of standardized model formats, and the dissemination of open-source tools, thereby promoting harmonized model performance evaluation criteria and clear pathways for regulatory acceptance^{125,207,251,296,303,304}. Transnational cooperation would not only expand data diversity and reduce model bias, but also facilitate mutual recognition and coordination among regulatory agencies across countries²⁰⁷. Naturally, open collaboration must also address challenges related to intellectual property and data sovereignty. By exploring innovative cooperation mechanisms, it is possible—while safeguarding the privacy of all parties' data—to realize the goal of “data sharing and collaborative modeling”^{232,270,271,281}. Open science and benchmarking are advancing rapidly. The 2025 Virtual Cell Challenge tests perturbation-response prediction on unseen cell types via open competition, improving method comparability and iteration¹⁴⁶. Therapeutic Data Commons (PyTDC) provides a unified evaluation framework by aggregating drug-discovery and single-cell benchmarks³⁰⁵. The “Open Problems in Single-Cell Analysis” initiative releases benchmark datasets and metrics to coordinate community solutions³⁰⁶. These efforts establish shared standards and reproducible resources, strengthen cross-dataset/platform external validity, and enable rapid, robust optimization of virtual cell models. These measures will cultivate a more open academic environment and accelerate the accumulation and iterative refinement of the underlying technologies^{2,307,308}.

Outlook

Virtual cell models are poised to evolve from auxiliary tools in preclinical research into a core component of the drug-evaluation framework^{1,6,12}. Their simulation scope will extend beyond current intracellular processes to higher organizational levels—including tissue morphogenesis and organ-level functional dynamics—thereby enabling a more complete depiction of complex physiological and pathophysiological processes^{309,310}. In addition, establishing internationally recognized benchmark datasets and quantitative

metrics will allow objective comparisons of different models' predictive performance and credibility^{1,16,251,311,312}. In parallel, explainable-AI approaches will be integrated into the development pipeline of virtual cell models to document and evaluate, end-to-end, the evidentiary basis of model decisions, improving the traceability of predictions and their clinical acceptability; as transparency increases and validation cases accumulate, clinical and regulatory confidence in these models is expected to strengthen progressively^{313–316}. Looking ahead, virtual cell technologies are expected to play a central role in drug discovery, disease modeling, and personalized medicine, achieving a closed loop from theoretical innovation to clinical translation¹². Driven by the explosive growth of biological big data, sustained advances in computing hardware, and broader societal acceptance of experimental alternatives, virtual cell models are likely to transition from frontier exploration to mainstream application in the near future, catalyzing a paradigm shift across the life sciences and the biopharmaceutical industry^{1,317} (Fig. 6).

Data availability

No datasets were generated or analyzed during the current study.

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During the preparation of this work, the authors used ChatGPT-5Auto for language polishing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Additional information

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